Methods Protocol: Governance, Provenance, and Validation

Event Logging and Participant-Facing Governance I

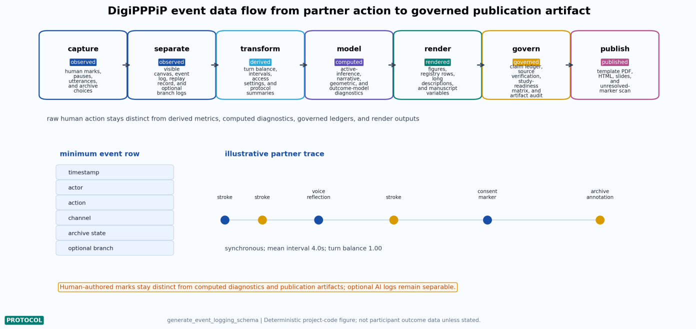
DigiPPPiP is presented as a framework manuscript, so its method layer is a protocol specification rather than a participant dataset. The protocol requires timestamped events with actor, action, and channel fields; the example trace in [@fig:event_logging_schema] has 6 summary fields, a mean interval of 4 seconds, and a turn-balance score of 1. This makes the distinction among synchronous, semisynchronous, and asynchronous sessions auditable instead of rhetorical. The same event stream supports replay, accessibility auditing, outcome modeling, and synchronization with optional physiological channels, but those uses must stay visibly downstream from the partner-facing mark.

Event Logging and Participant-Facing Governance II

Digital pen and paper research in observational settings gives a practical precedent for treating pen traces as analyzable field records rather than merely as drawings [@weibel2012digitalpen]. DigiPPPiP extends that idea to a dyadic protocol: the stroke is simultaneously a mark for the partner, an event for replay, and a data row for later audit. That dual role is why the manuscript separates the visible canvas from the event log, source ledger, and generated figure registry.

[@fig:event_logging_schema] now treats the method layer as an end-to-end data-flow contract. Raw partner actions are captured as consented event rows, separated from the visible canvas and replay record, transformed into summaries, optionally modeled, rendered as figures and variables, governed by claim/source/readiness audits, and finally published through the template pipeline. The important architectural rule is separation: human-authored marks, optional AI outputs, computed diagnostics, and publication artifacts should remain distinguishable in both the protocol and the figures.

Event Logging and Participant-Facing Governance III



Event Logging and Participant-Facing Governance IV

Figure 1: Event schema, data-flow path, and example DigiPPPiP session trace. The figure is generated by `generate_event_logging_schema()` from `src/session_events.py` and `src/systems_governance.py`; read the top row as the capture-to-publication path, the lower-left list as minimum event-row fields, and the timeline as an illustrative actor/action sequence with temporal summary metrics. It supports the protocol claim that synchrony categories, model diagnostics, generated figures, and publication artifacts must remain auditable. The trace is illustrative, and participant studies would replace it with consented logs while keeping optional AI and physiology logs separable.

Event Logging and Participant-Facing Governance V

Before human-subjects work, the protocol needs a participant-facing layer as well as an analysis layer. The Common Rule and the Declaration of Helsinki require consent, review, privacy, protocol clarity, withdrawal, and risk handling to be explicit rather than inferred from good intentions [@hhs2025cfr46;@wma2024helsinki]. Digital-health consent scholarship adds that participant understanding depends on concrete information about access, purpose, privacy protection, and oversight [@kassam2023digitalconsent]. The plain-language brief should say that two partners are invited to draw on a shared surface; that the study records timing, tool, archive-control, and optional replay events; that the activity is not treatment or a relationship test; and that participants can pause, stop, withdraw, ask questions, report discomfort, request redaction, and ask about save, replay, export, and delete options before analysis lock. If one partner wants more sharing and the other wants less, the stricter privacy choice governs. This participant-facing appendix is generated as a protocol artifact by `src/study_readiness.py` rather than left as informal prose.

Source and Claim Governance I

The study protocol separates source quality from claim strength. Peer-reviewed empirical and review articles can support scoped design and measurement claims; books support theoretical framing; preprints and reports are treated as provisional. [@fig:source_quality_map] encodes those rules so speculative additions, including geometric hyperscanning and narrative-information proposals, are not used as settled evidence. This is why digital object identifier (DOI) verification matters: a search-result citation is not added unless title, authors, venue, and DOI or stable uniform resource locator (URL) match the intended claim. The stricter rule is claim-domain specific: therapy, neural synchrony, active inference, accessibility, placemaking, digital intimacy, AI mediation, and privacy persistence each have a different maximum defensible claim strength before new evidence is collected.

Source and Claim Governance II

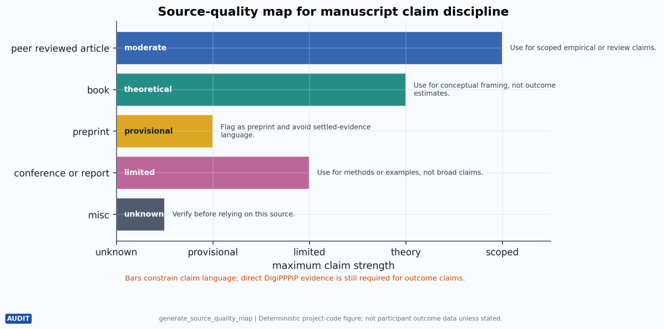


Figure 3: Source-quality map for DigiPPPiP claim discipline. The figure is generated by `generate_source_quality_map()` from `src/source_quality.py`; read the horizontal bars as conservative maximum claim strength by source class, the embedded labels as source-tier roles, and the right-hand text as warnings for manuscript use. It supports the protocol rule that bibliography type constrains prose strength. The map is an audit device, not a judgment of individual papers, and claims can strengthen only with domain-appropriate empirical evidence.

Source and Claim Governance III

[@fig:claim_boundary_matrix] makes that red-team layer explicit across 14 recurring claim domains. The point is not to prohibit ambitious studies; it is to keep the manuscript from borrowing strength across domains. A valid shared-workspace source can motivate the interface, but it cannot establish relationship improvement. A hyperscanning method source can motivate instrumentation, but it cannot establish causality or therapeutic benefit. A digital-art-therapy review can motivate caution and study design, but it cannot turn DigiPPPiP into treatment without direct evidence.

Source and Claim Governance IV

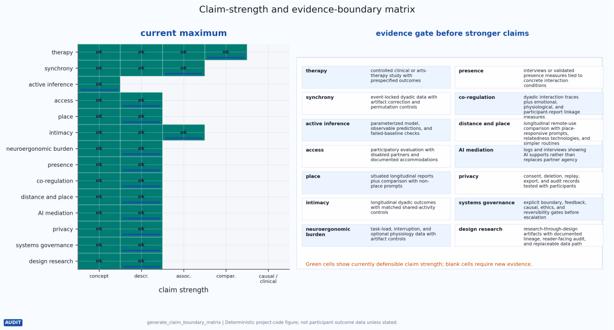


Figure 4: Claim-strength and evidence-boundary matrix for DigiPPPiP. The matrix is generated by `generate_claim_boundary_matrix()` from `src/source_quality.py`; read rows as reviewer-risk domains, labelled green cells and blue rule marks as currently defensible claim levels, and the right panel as the evidence gate required for stronger language. It supports adversarial validity checking. The matrix is conceptual, and a domain can move only after direct study evidence is added.

Source and Claim Governance V

The boundary matrix is paired with a claim ledger rather than left as an abstract warning. `src/claim_ledger.py` records recurring manuscript claim families, their section, citekeys, present maximum strength, and the evidence needed to upgrade them. [@fig:claim_ledger_matrix] makes the ledger visible. This also documents the role of Perplexity and web search in the workflow: discovery tools may nominate sources, but a citekey enters the ledger only after the title, venue, year, and DOI or stable URL have been checked against Crossref, a publisher page, an official standard, or an archival record.

Source and Claim Governance VI

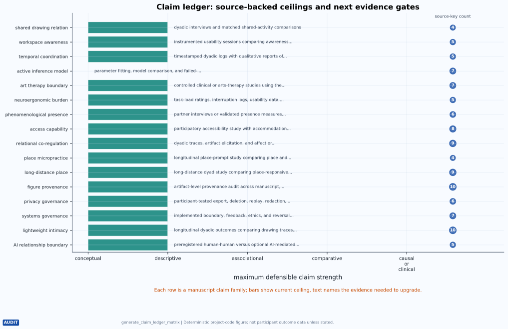


Figure 5: Claim-ledger matrix linking manuscript claim families to evidence ceilings and upgrade gates. The figure is generated by `generate_claim_ledger_matrix()` from `src/claim_ledger.py`; read each row as a stable claim family, each bar as the current maximum claim strength, the blue numbered marker as evidence-key count, and the right-hand text as the next evidence gate. It supports source-to-claim governance rather than an empirical result. A stronger bar requires direct study evidence, not a more adjacent citation.

Source and Claim Governance VII

`src/source_verification.py` makes that rule executable. Each governed citekey receives a locator, verification URL, checked-as-of date, source tier, claim family, manuscript location, and recheck trigger. The audit prioritizes 2024–2026 sources, preprints, AI claims, digital-health claims, governance sources, and reporting standards for refresh before submission. [@fig:source_verification_readiness] renders the current source-verification state from the same governed citekey surfaces used by tests: claim ledger, evidence graph, figure-method source bridge, and study-readiness cases. The ledger is intentionally stricter than a bibliography because it asks what claim family a source is allowed to support and when it must be rechecked.

Source and Claim Governance VIII

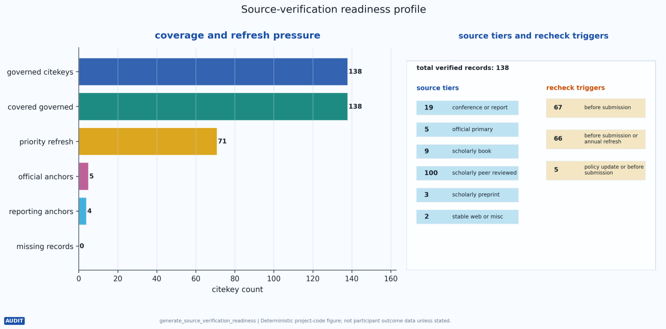


Figure 6: Source-verification readiness profile for governed DigiPPPiP citekeys. The figure is generated by `generate_source_verification_readiness()` from `src/source_verification.py`; read the left panel as labelled citekey coverage and priority-refresh pressure, and the right panel as source-tier and recheck-trigger counts in audit cards. It supports provenance auditing rather than literature synthesis. A zero missing-record count means the ledger has locators and metadata, not that every source independently validates DigiPPPiP.

Systems Boundary and Feedback Governance I

A first-principles pass reduces DigiPPPiP to a small hard kernel: two human partners, perceivable marks, temporal structure, response opportunity, and accountable persistence. `src/systems_governance.py` makes the rest explicit as governed branches rather than implicit scope creep. The current systems layer records 6 boundary elements, 5 feedback loops, 5 causal assumptions, and 5 ethics gates, with a completeness score of 1. Active-inference and enactive accounts support the action-perception framing and Markov-blanket caution, while contextual privacy, values-in-design, and human-subjects governance anchor the participant-facing side of the boundary [@ramstead2020two;@nissenbaum2011contextualprivacy;@shilton2012values;@hhs2025cfr46;@wma2024helsinki].

Systems Boundary and Feedback Governance II

The boundary table keeps the human-human drawing loop inside the kernel and marks event logs, place context, physiology, AI assistance, and clinical translation as support structures or optional branches. Each branch has a reversal gate: place prompts can be declined or coarsened; physiology requires separate consent and artifact controls; AI suggestions must be labelled, rejectable, undoable, and separable from partner authorship; clinical translation stays out of scope until a reviewed intervention protocol exists. This is a systems-governance claim, not an efficacy claim. It says that the framework can name its boundaries and stop conditions before it asks participants or reviewers to trust them.

Systems Boundary and Feedback Governance III

The feedback layer is deliberately adversarial. Action-perception loops should preserve mutual agency; privacy loops should default to the stricter dyadic archive choice; access loops should respond to fatigue, sensory mismatch, and assistance needs; AI loops should detect authorship confusion or partner substitution; evidence-escalation loops should downgrade language when controls or participant reports do not support a stronger claim. This makes falsification operational: if matched controls explain the same relatedness reports, if place prompts do not change situated memory, if replay feels like surveillance, if AI reduces co-authorship, or if physiological linkage disappears after artifact correction, the corresponding claim boundary narrows rather than hardens.

Evidence Scope and Non-Claims I

The present manuscript implements a design and governance architecture, not a completed empirical evaluation. It demonstrates that DigiPPPiP concepts can be expressed as manuscript sections, tested primitives, generated figures, source ledgers, claim boundaries, protocol gates, and render artifacts. It does not demonstrate therapeutic efficacy, relationship improvement, neural coupling, universal accessibility, or AI benefit. Active inference, geometric hyperscanning, narrative information, and outcome modeling are used as modeling and measurement lenses; they become mechanisms only if future studies fit them to observed sessions, compare them with plausible alternatives, and survive the falsifiers named in the protocol.

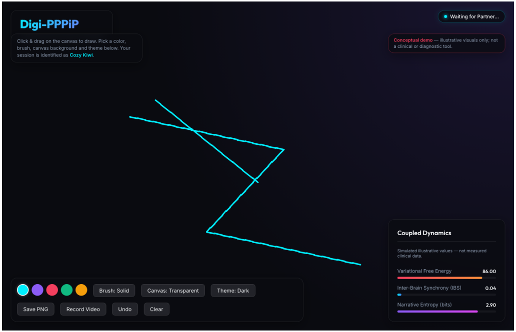
Evidence Scope and Non-Claims II

This scope statement also governs visual interpretation. Architecture figures show boundaries, not deployed software. Data-flow figures show what a study should record, not what participants have already produced. Simulation plots show toy diagnostics, not human behavior. Audit figures show that the manuscript has local governance machinery, not that DigiPPPiP is externally validated. Those distinctions are intentionally repetitive because they are the main defense against attractive diagrams becoming overclaims.

Interactive Web Instantiation (Conceptual Demo) I

DigiPPPiP is also implemented as a small, self-contained interactive web canvas so that the cyberphysical argument in [@sec:cyberphysical] is concrete rather than diagrammatic only. The application pairs people on a shared, low-latency drawing surface and relays strokes, undos, and cursor positions through a Socket.IO server; each participant is identified by a lightweight Adjective-Fruit moniker that follows their cursor across the shared field. Its design kernel is deliberately the minimum viable kernel described in [@sec:cyberphysical]: partners, a shared mark field, perceptible traces of agency, and consentful control over persistence. The running tool is therefore a working instantiation of the architecture in [@fig:cpss_architecture] rather than a clinical instrument.

Interactive Web Instantiation (Conceptual Demo) II



Interactive Web Instantiation (Conceptual Demo) III

Figure 7: The running Digi-PPPiP web canvas in its default dark theme, showing freehand strokes, the active-user moniker (Cozy Kiwi), the waiting-for-partner status, the drawing palette, and the simulated Coupled Dynamics panel. Captured from the running Vite/Socket.IO demo stack by `generate_webapp_main_canvas()`; read the canvas as the shared mark field, the top bar as session state, and the bottom-right panel as the computed modeling lens. It supports the methods argument that the cyberphysical substrate in [@fig:cpss_architecture] is implementable. The screenshot is conceptual; assessed usability and relational effects would require the study protocol in [@sec:methods_prot ocol].

Interactive Web Instantiation (Conceptual Demo) IV

The screen captures deliberately confine the relational claim. [@fig:webapp_main_canvas] shows a single-user session; the paired dynamics the framework theorizes are rendered only as simulated values. [@fig:webapp_metrics_dashboard] isolates the live dashboard, where Variational Free Energy, Inter-Brain Synchrony, and Narrative Entropy are displayed with an explicit simulated-illustrative-values disclaimer. These numbers are generated client-side as a demonstration of the modeling lens; they are not experimental measurements and must not be read as evidence of neural coupling or therapeutic effect.

Interactive Web Instantiation (Conceptual Demo) V

Interactive Web Instantiation (Conceptual Demo) VI

Figure 8: The Coupled Dynamics dashboard as rendered in the live app, isolating the simulated Variational Free Energy, Inter-Brain Synchrony, and Narrative Entropy readouts and their progress bars. Captured by `generate_webapp_metrics_dashboard()`; read each row as metric label, value, and relative magnitude. It supports the protocol distinction between a computed modeling lens and an outcome measurement. The values are simulated and illustrative; real values would require physiological instrumentation and the consent and logging gates in [@sec:methods_protocol].

Interactive Web Instantiation (Conceptual Demo) VII

The interface also exposes configuration choices that matter for the accessibility and privacy commitments in [@sec:accessibility]. [@fig:webapp_light_theme] shows the same canvas in a light theme, and [@fig:webapp_theme_settings] shows the theme and canvas-background settings against a plain white canvas. These options are not cosmetic polish; they operationalize the ability-based and WCAG-informed design commitments of the framework by letting a pair choose contrast and background settings that fit their access needs, while session and archive controls remain distinct from the visible drawing surface.

Interactive Web Instantiation (Conceptual Demo) VIII

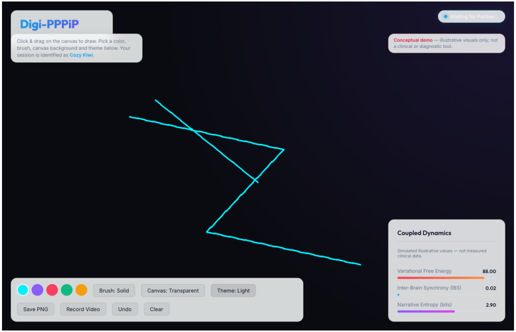


Figure 9: The same Digi-PPPiP canvas switched to the light theme. Captured by `generate_webapp_light_theme()`; read it as the tool's visual-contrast alternative to [@fig:webapp_main_canvas]. It supports the accessibility commitment that theme and contrast are user-selectable. The screenshot is conceptual; demonstrated accessibility benefit would require an accessibility audit such as [@fig:accessibility_audit_radar].

Interactive Web Instantiation (Conceptual Demo) IX

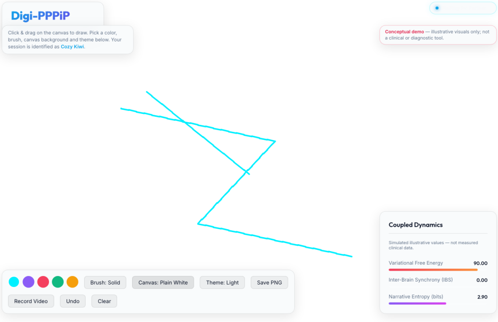


Figure 10: Theme and canvas-background settings shown in the light theme against a plain white canvas. Captured by `generate_webapp_theme_settings()`; read the control row as the selectable theme, brush, and background configuration. It supports the design-commitment argument that visual-contrast and background choices are first-order, user-facing settings. The screenshot is conceptual and shows interface configuration, not measured accessibility outcomes.

Reproducible Figure Methods I

Reproducible Figure Methods II

The figure-generation method is also part of the research protocol rather than a cosmetic afterthought. Visualization grammars clarify how data, marks, scales, and layers compose [@wilkinson2005grammar;@wickham2010layered;@bostock2011d3;@satyanarayan2017vegalite], while graphical-perception, color-use, and validation work shows that scientific figures need perceptual, task, and design justification rather than merely attractive styling [@cleveland1984graphical;@munzner2009nested;@crameri2020colour]. DigiPPPiP therefore treats each generated figure as a small reproducible claim object: a scoped argument, a method-lineage warrant, a tested primitive or explicitly conceptual source, a visual encoding grammar, an aesthetic/accessibility rule, a deterministic render, a registry entry, a caption contract, an accessibility description, and a render validation gate. [@fig:figure_generation_pipeline] encodes that artifact chain. The implemented method has 9 generation stages, 7 semantic encoding roles, 5 source-method families, 13 audit criteria, 8 required caption elements, and a method-contract score of 1. The relevant scholarship supports this restraint: effective figures need explicit design choices and accessible

Study Package, Access, Place, and Outcomes I

The minimum reproducibility package for a DigiPPPiP study should include the protocol version, temporal-spatial condition, interface capabilities, event schema, exclusion windows, consent and persistence settings, accessibility accommodations, outcome measures, and analysis code. Without that package, readers cannot tell whether an observed result belongs to the dyad, the task, the technology, the access arrangement, or the analysis pipeline. This is a first-principles requirement: if the causal candidates cannot be separated, the interpretation cannot be audited.

Study Package, Access, Place, and Outcomes II

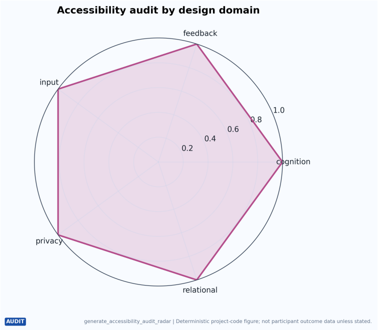
The protocol appendix should also separate three participant-facing surfaces that are easy to collapse. The access surface records supported input channels, feedback channels, partner-mediated assistance, cognitive-load settings, communication preferences, and accommodation changes during the session. The place surface records only the situated cues that participants consent to share, such as room type, local prompt, mobility context, or remembered place, and it should not require precise geolocation for a place-responsive claim. The health surface states that DigiPPPiP is non-clinical unless a reviewed intervention protocol says otherwise, names stop and referral procedures, and separates personal keepsakes from research logs before any export or replay.

Study Package, Access, Place, and Outcomes III

Reporting guidance supplies the study-readiness bridge. The Template for Intervention Description and Replication (TIDieR) requires enough intervention detail for replication, and the Consolidated Standards of Reporting Trials for eHealth (CONSORT-EHEALTH) names web and mobile health reporting elements that are easy to miss when an online intervention feels self-explanatory [@hoffmann2014tidier;@eysenbach2011consortehealth]. For DigiPPPiP that means the protocol must define session dose, materials, platform settings, tailoring, fidelity, archive-control states, and changes made during the study. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) is not activated here because the evidence graph is a bounded narrative synthesis rather than a systematic review claim; it should be added only if the review question, search strategy, screening, extraction, and flow diagram become systematic.

Study Package, Access, Place, and Outcomes IV

Accessibility is handled as a protocol requirement. The current audit has 5 criteria and an illustrative implementation score of 1. The radar in [@fig:accessibility_audit_radar] is not a compliance certificate; it is a reproducible way to document which input, feedback, privacy, cognitive-load, and partner-mediation capabilities a study or tool actually supports.



Study Package, Access, Place, and Outcomes V

Figure 15: Accessibility audit radar generated from explicit implementation capabilities. The figure is generated by `generate_accessibility_audit_radar()` from `src/accessibility.py`; read each spoke as a capability domain and radial distance as documented support in the illustrative configuration. It supports reproducible access auditing, not compliance certification. Empirical accessibility claims would require participatory validation with disabled users and accommodation logs.

Outcomes are organized as a multilevel design rather than a single success metric. The project defines 6 measures across 6 domains, with a default design-strength score of 4 for a randomized longitudinal study. [@fig:multilevel_outcome_model] shows how temporal mode, spatial configuration, phase, and access condition can be modeled with dyad and participant random effects. This structure separates implementation outcomes, such as latency and logging completeness, from relational outcomes, such as perceived connection and shared meaning.

Study Package, Access, Place, and Outcomes VI

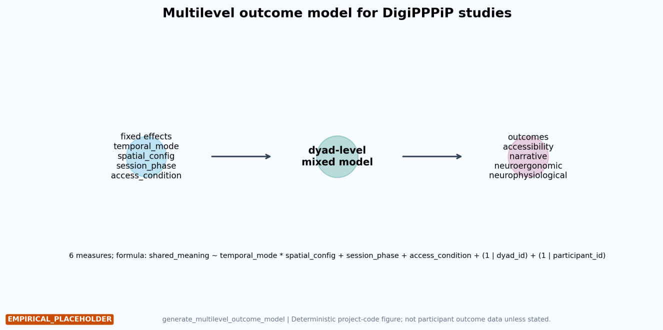


Figure 16: Multilevel outcome model for DigiPPPiP studies. The figure is generated by `generate_multilevel_outcome_model()` from `src/outcomes.py`; read left components as fixed effects, the center as the dyad-level mixed model, and the right component as outcome domains. It supports planned-analysis specification rather than reporting results. Validation would require real dyadic observations, random effects diagnostics, and matched shared-activity controls.

Falsification and Claim Boundaries I

Falsification should be designed in. The framework would be weakened if matched controls produce the same relational outcomes, if accessibility accommodations reduce rather than expand agency, if synchrony measures vanish after artifact correction, if AI suggestions displace rather than support human agency, or if participants describe persistent archives as surveillance rather than shared memory. Those outcomes would not invalidate shared drawing as a practice; they would delimit which mechanisms, contexts, or claims the framework can support.

[@fig:validation_ladder] expresses this as 5 staged gates. Feasibility and relational meaning come before access claims, matched outcome comparisons, and optional physiology. This ordering is adversarial by design: neural or computational sophistication cannot rescue a practice that is not understood, consentful, accessible to the stated population, or meaningfully different from simpler shared activities.

Falsification and Claim Boundaries II

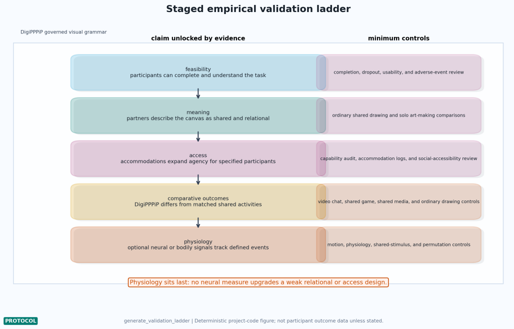


Figure 17: Staged empirical validation ladder for DigiPPPiP. The figure is generated by `generate_validation_ladder()` from `src/source_quality.py`; read the left column as claims unlocked by each stage and the right column as minimum controls before escalation. It supports the protocol's adversarial ordering from feasibility and meaning through access, matched outcomes, and optional physiology. The ladder is conceptual, and stronger claims require completed studies at the relevant stage.

Falsification and Claim Boundaries III

This protocol prevents the main evidentiary errors. Active inference is used as a modeling language, not proof of mechanism. Hyperscanning synchrony is treated as a correlate until causal designs justify more [@cui2012nirs;@czeszumski2020hyperscanning; @hamilton2021hyperscanning;@zimmermann2024methodological]. Art-therapy and digital-health literatures motivate possible applications and cautions, but DigiPPPiP is not described as treatment without controlled clinical evidence [@zubala2021digitalarttherapy]. Accessibility and placemaking claims stay bounded to the capabilities and populations actually tested.